

Impacts of Major Storm Flooding on Harbor Water Quality in New York City using Time Series and Spatial Analysis

Carvan, S.C.¹, Carvan, Z.C.¹, and Marsellos, A.E.¹

¹Dept. Geology, Environment and Sustainability, Hofstra University, Hempstead, NY, USA



Abstract:

Water quality degradation following major storm events poses significant public health risks in urbanized coastal environments. This study examines the impacts of four storm events on bacterial and chemical contaminant levels in New York City Harbor using water quality data from the NYC Department of Environmental Protection Harbor Water Quality Survey (1909–2024). The storms analyzed—Hurricane Sandy (2012), Tropical Storm Fay (2020), Hurricane Ida (2021), and Tropical Storm Ophelia (2023)—represent different storm types and flooding mechanisms. A ± 14 -day window around each landfall was used with Wilcoxon rank-sum tests to compare pre- and post-storm concentrations of Enterococcus, Fecal Coliform, Turbidity, and Nitrate. Station-level spatial maps were created using Inverse Distance Weighted interpolation to visualize contaminant distribution across the harbor. Storm flooding mechanism, not intensity alone, is the primary driver of water quality response. Hurricane Sandy, a surge-dominated event, produced significant post-storm bacterial increases (814% Enterococcus, 236% Fecal Coliform), with Spearman correlation showing contamination rising for days after landfall ($\rho = 0.62$, $p = 0.0003$), suggesting prolonged sewage infrastructure damage. Rainfall-dominated storms—Hurricane Ida, Tropical Storm Fay, and Tropical Storm Ophelia—produced dilution effects or no significant change, despite Ida being Category 4. These findings demonstrate that flood mechanism is a stronger predictor of harbor water quality response than storm intensity, with implications for post-storm public health monitoring and coastal stormwater management.

Background Information:

- New York City Harbor receives runoff from a heavily urbanized watershed with aging combined sewer infrastructure, making it vulnerable to contamination during storm events.
- Hurricane Sandy resulted in 11 billion gallons of untreated and partially treated sewage entering regional waterways due to record storm surge overwhelming treatment facilities (Climate Central, 2013).
- As climate change intensifies both storm surge and rainfall events, understanding their distinct impacts on coastal water quality is increasingly critical.
- Previous studies have focused on storm intensity as a predictor of water quality degradation, but the role of flooding mechanism, whether surge-driven or rainfall-dominated, remains poorly understood.

Method/Materials:

- NYC DEP Harbor Water Quality Survey (1909–2024)
 - Weekly/monthly sampling across monitoring stations
 - Contaminants: Enterococcus, Fecal Coliform, Turbidity, Nitrate
- Storm Events
 - Hurricane Sandy (2012, surge-dominated)
 - Tropical Storm Fay (2020, rainfall-dominated)
 - Hurricane Ida (2021, rainfall-dominated, Category 4)
 - Tropical Storm Ophelia (2023, rainfall-dominated)
- RStudio Analysis
 - ± 14 -day window around storm landfall
 - Wilcoxon rank-sum tests ($p < 0.05$)
 - Spearman rank correlation (temporal trends)
 - Percentage change by flooding mechanism
- Spatial Mapping
 - Inverse Distance Weighted interpolation
 - Station-level before/after contaminant maps

Results:

- Wilcoxon rank-sum tests: 8 of 16 comparisons significant ($p < 0.05$)
 - Hurricane Sandy (Surge-Dominated)
 - Enterococcus: +814% ($p = 0.0014$)
 - Fecal Coliform: +236% ($p = 0.038$)
 - Spearman correlation showed prolonged contamination days after landfall ($\rho = 0.62$, $p = 0.0003$)
- Rainfall-Dominated Storms (Fay, Ida, Ophelia)
 - Fay: Enterococcus increased ($p = 0.0009$)
 - Ida: Fecal Coliform decreased 43% ($p = 0.00008$) despite Category 4 intensity
 - Ophelia: Enterococcus decreased 87% ($p = 0.0000005$), Fecal Coliform decreased 69% ($p = 0.011$)
- Key Finding
 - Flooding mechanism (surge vs. rainfall) is the primary driver of water quality response, not storm intensity
 - Station-level spatial maps revealed localized contamination hotspots across the harbor

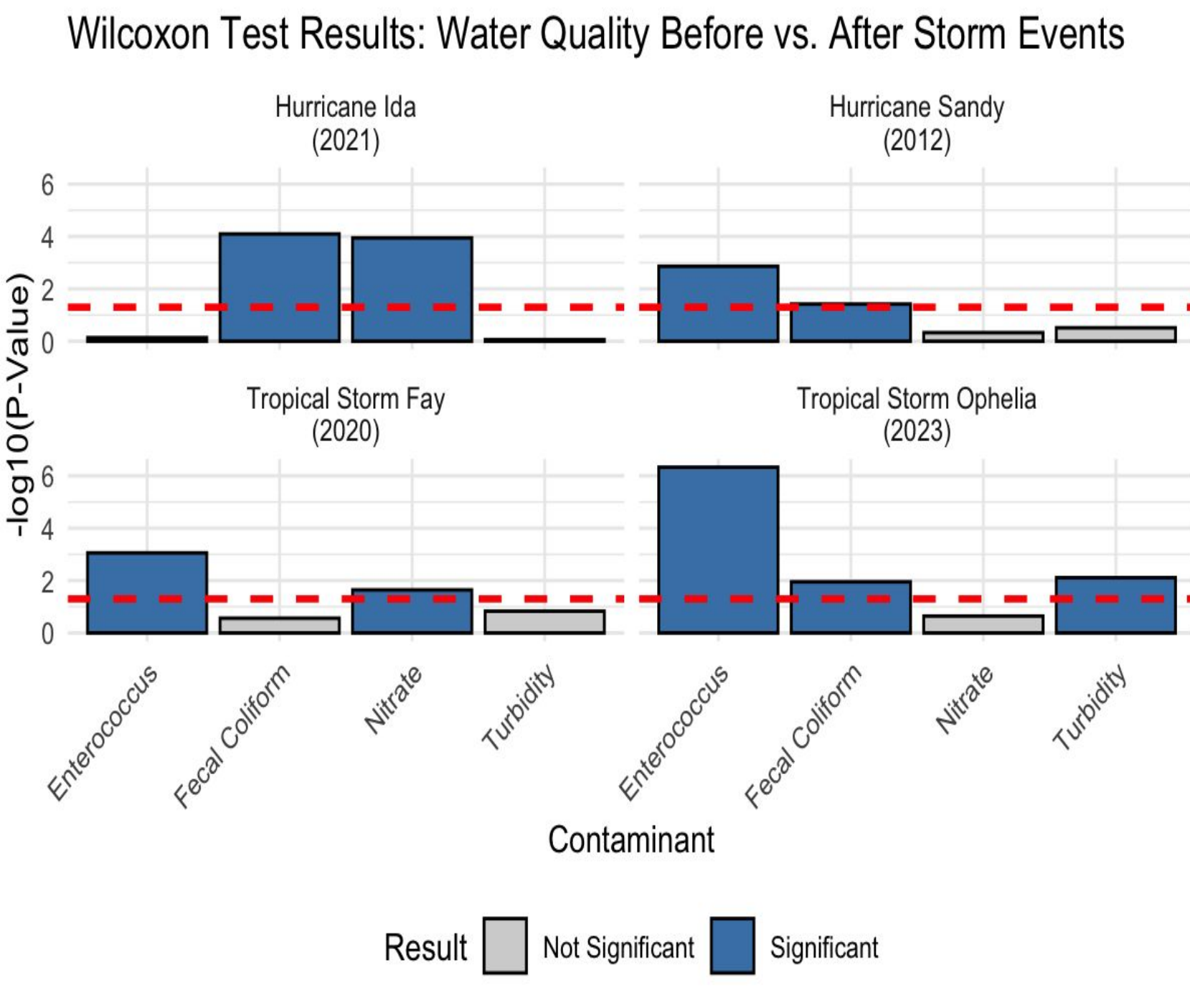


Figure 1: Wilcoxon rank-sum test results showing statistical significance of water quality changes across four storms. Blue bars indicate significant results ($p < 0.05$); gray bars indicate non-significant results. 8 of 16 comparisons reached significance.

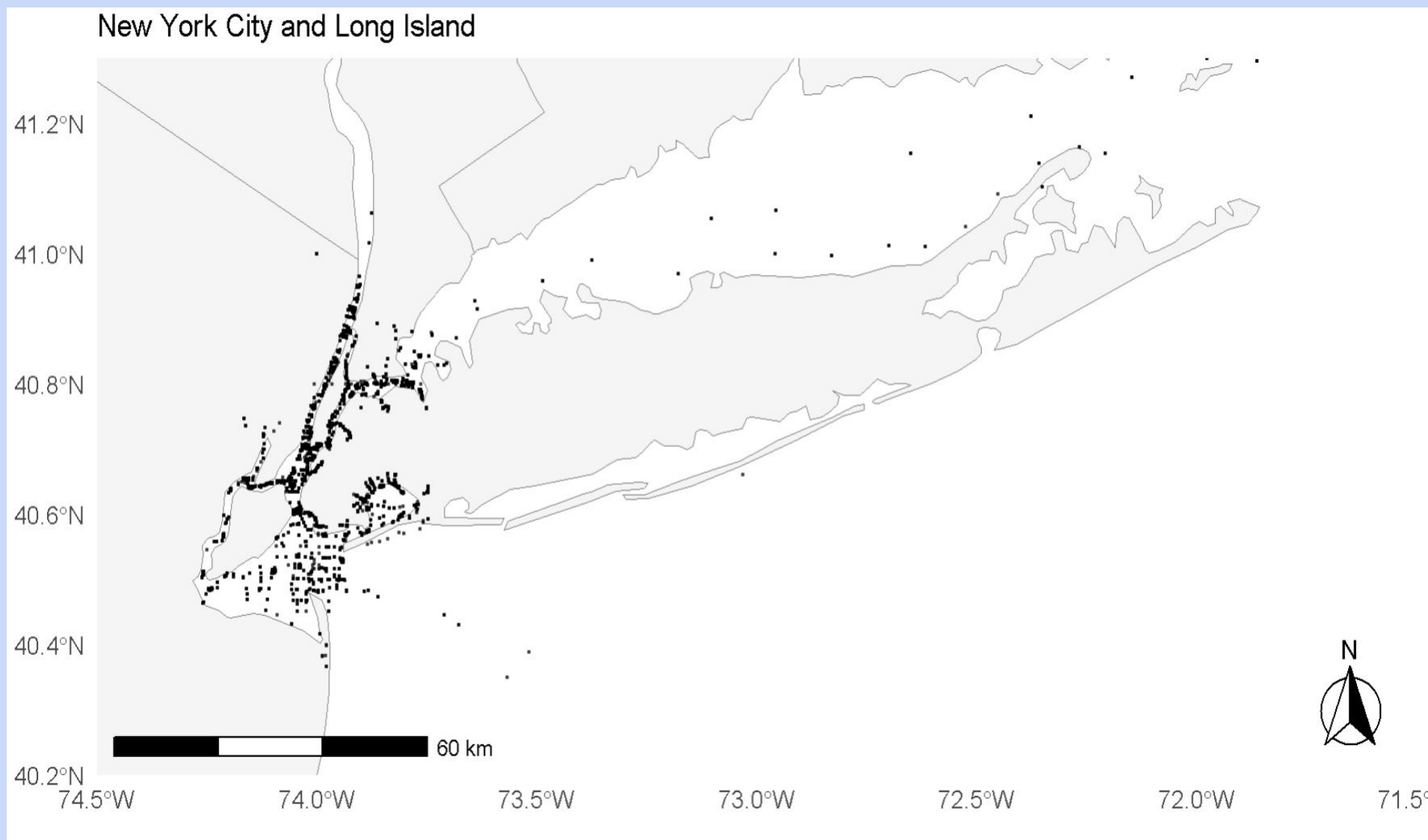


Figure 2: Sampling Locations: NYC DEP monitoring stations across New York City Harbor and surrounding waters. Black dots indicate station locations used in water quality analysis.

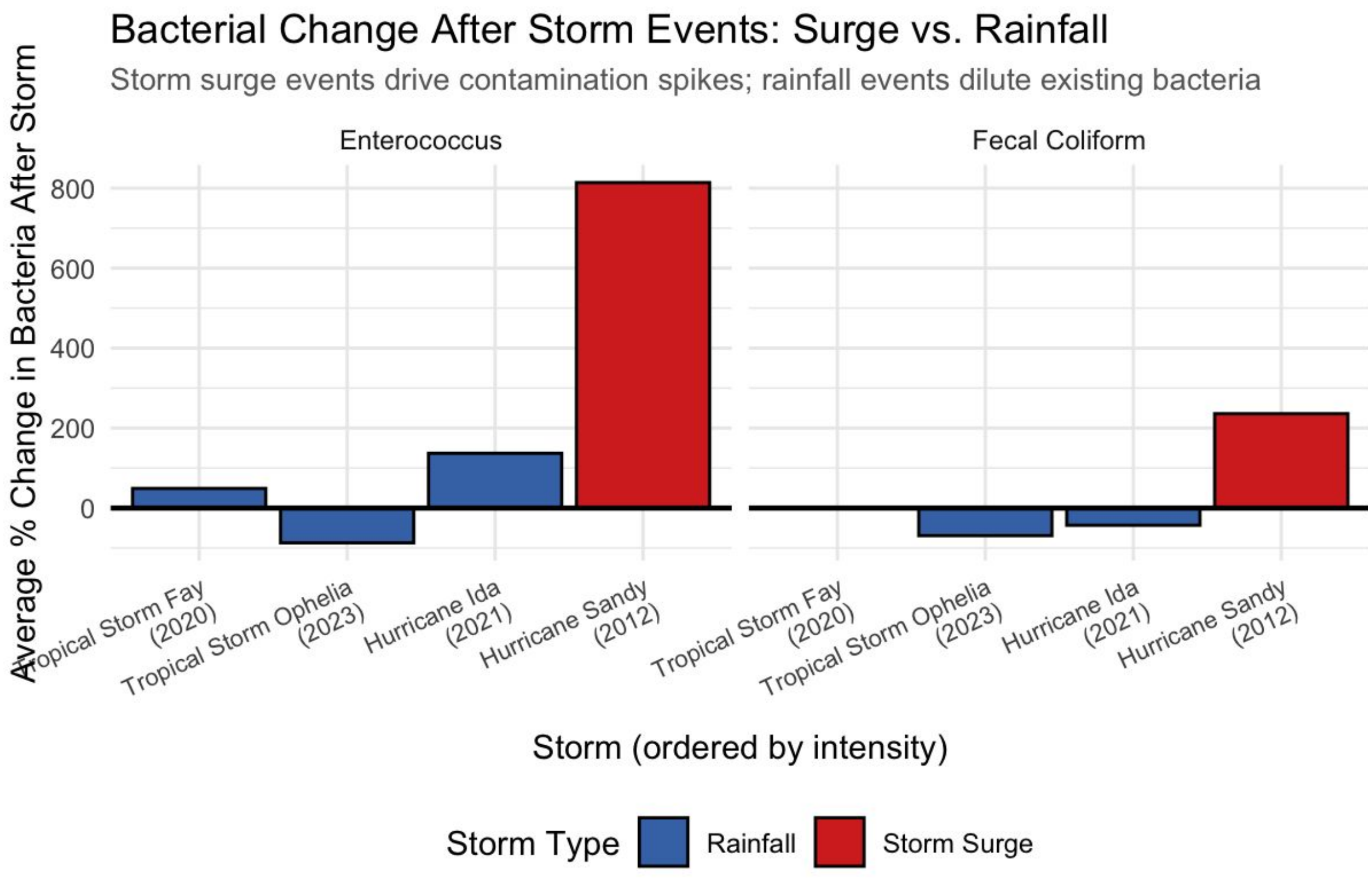


Figure 3: Average percentage change in Enterococcus and Fecal Coliform concentrations post-storm compared to pre-storm, grouped by flooding mechanism. Red bars indicate surge-dominated Hurricane Sandy; blue bars indicate rainfall-dominated storms.

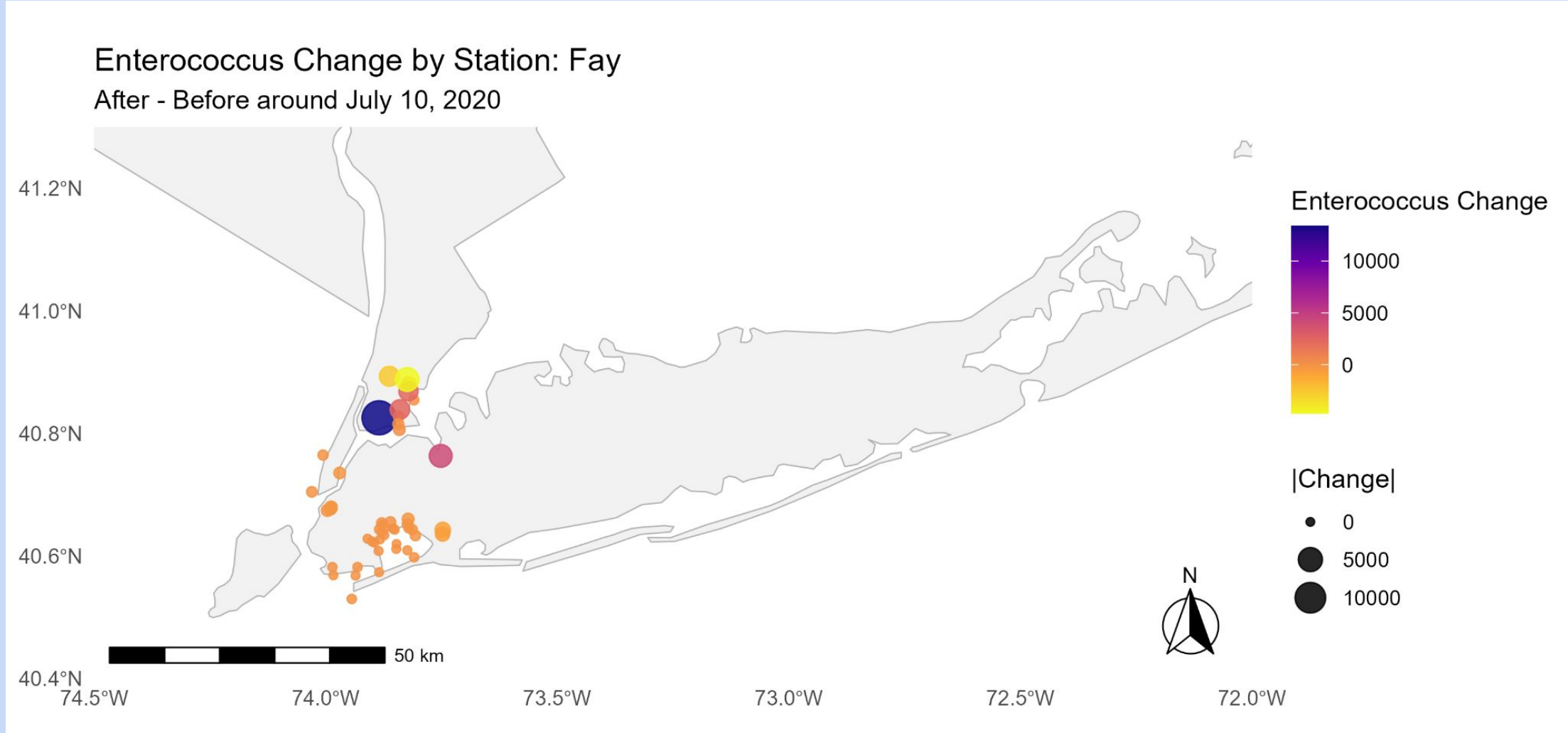


Figure 4: Station-level Enterococcus change map for Tropical Storm Fay using Inverse Distance Weighted interpolation. Warmer colors indicate increases; cooler colors indicate decreases. Dot size reflects magnitude of change.

Discussion/Conclusion:

- Storm flooding mechanism, not storm intensity, was the primary driver of water quality change in NYC Harbor
- Hurricane Sandy (surge-driven) produced the largest bacterial increases: Enterococcus +814%, Fecal Coliform +236%
- Spearman correlation confirmed contamination continued rising after Sandy ($\rho = 0.62$, $p = 0.0003$), suggesting prolonged sewage system damage
- All three rainfall-dominated storms showed bacterial dilution or no significant change, including Category 4 Hurricane Ida
- Averaging across all stations removes spatial variability and may mask localized contamination near flood-prone areas
- Future analysis will incorporate FEMA flood zone designations at the individual station level for a more spatially precise assessment

Acknowledgments:

The authors acknowledge Dr. Antonios E. Marsellos and his Geoinformatics Laboratory at Hofstra University for fostering an interdisciplinary research environment that integrates geoinformatics, artificial intelligence, and geohazard analysis, enabling the successful completion of this study.

Credit Authorship Contribution Statement:

Carvan, S.: Data collection, statistical analysis, RStudio coding, writing of Abstract, Introduction, Methods, Results, Discussion, Conclusion, and Figures 1, 2, 3; editing.

Carvan, Z.: Data collection, statistical analysis, writing of Methods, Results, and Discussion; data analysis, editing, and contribution to Figure 2.

Marsellos, A.E.: Project supervision, RStudio coding, spatial analysis, review and revision of manuscript, and contribution to Figure 3.

References:

- Becker, R.A., Wilks, A.R., Brownrigg, R., Minka, T.P., and Deckmyn, A. (2025). maps: Draw Geographical Maps. R package version 3.4.3. <https://CRAN.R-project.org/package=maps>
- Dunington, D. (2025). ggspatial: Spatial Data Framework for ggplot2. R package version 1.1.10. <https://CRAN.R-project.org/package=ggspatial>
- FEMA Map Service Center National Flood Hazard Layer. <https://msc.fema.gov>
- Garner, S., Ross, N., Rudis, R., Camargo, A.P., Scaini, M., and Scherer, C. (2024). viridis: Colorblind-Friendly Color Maps for R. R package version 0.6.5. <https://doi.org/10.5281/zenodo.4679423>
- Graler, B., Pebesma, E. and Heuvelink, G. (2016). Spatio-Temporal Interpolation using gstat. The R Journal, 8(1), 204-218.
- Massicotte, P. and South, A. (2026). naturalearth: World Map Data from Natural Earth. R package version 1.2.0. <https://CRAN.R-project.org/package=naturalearth>
- NYC Department of Environmental Protection. Harbor Water Quality Survey. NYC Open Data. https://data.cityofnewyork.us/Environment/Harbor-Water-Quality/Suue-f89n/about_data
- Pebesma, E. (2018). Simple Features for R: Standardized Support for Spatial Vector Data. The R Journal, 10(1), 439-446. <https://doi.org/10.32614/RJ-2018-009>
- Pebesma, E. and Bivand, R. (2005). Classes and Methods for Spatial Data in R. R News, 5(2), 9-13. <https://cran.r-project.org/doc/Rnews/>
- Pebesma, E. and Bivand, R. (2023). Spatial Data Science: With Applications in R. Chapman and Hall/CRC. <https://doi.org/10.1201/9780429459016>
- Pebesma, E.J. (2004). Multivariable geostatistics in S: the gstat package. Computers and Geosciences, 30, 683-691.
- R Core Team (2024). R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria. <https://www.R-project.org/>
- Wickham, H. (2016). ggplot2: Elegant Graphics for Data Analysis. Springer-Verlag New York. <https://ggplot2.tidyverse.org>
- Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L.D., Francois, R., Grolemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T.L., Miller, E., Bache, S.M., Müller, K., Ooms, J., Robinson, D., Seidel, D.P., Spinu, V., Takahashi, K., Vaughan, D., Wilke, C., Woo, K., and Yutani, H. (2019). Welcome to the tidyverse. Journal of Open Source Software, 4(43), 1686. <https://doi.org/10.21105/joss.01686>